

ROVER LAB, DOSSIER (55 Cancr e mission)

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Planet

Name: 55 Cancr e ('Janssen')

Planet type: Ultra-hot rocky super-Earth with a permanent global lava ocean

Surface temperature: $\approx 1700\text{--}2400\text{ }^{\circ}\text{C}$ on the permanently lit dayside (JWST-measured; varies between observations, possibly due to volcanic outgassing or clouds). Tidally locked.

Surface pressure: Not directly measured. A real, volatile-rich secondary atmosphere (CO/CO₂-dominated) was confirmed by JWST in 2024.

Gravity: $\approx 2.3\text{ g}$

Atmosphere: CO- and CO₂-dominated, likely sustained by outgassing from the molten surface (JWST, Hu et al. 2024)

Mission Briefing

55 Cancr e, nicknamed 'Janssen', is a scorched rocky world orbiting so close to its Sun-like star that its dayside is a permanent ocean of molten rock. The rover 'Solidus-1' has been lowered onto a rare patch of solidified crust near the retreating edge of a lava flow, at the boundary between day and night. Cooling volcanic gases have left behind a bright orange-yellow crust rich in chloride salts. Mission control suspects this evaporite deposit, formed as superheated brines vented from cracks in the crust and flash-cooled, could be unusually concentrated in dissolved iron, exactly the kind of sample needed to produce iron oxide for radiation-shielding plates on the crew module still in orbit. Guide the rover to sample the deposit before its brief window on the twilight terminator closes.

Rover

Starting energy: 100

Site visit cost: 10

Experiment cost: 15

Overview Image

Site: Bright orange-yellow crust at the day-night terminator

Description: A crust of fused, glassy salts rimming a cooling lava flow, glowing faintly orange in the dim terminator light.

Identity: Concentrated iron(III) chloride solution (real sample, suggested ≈ 0.8 M), extreme dayside evaporation concentrates dissolved salts to unusually high levels. REAL SAMPLE GROUP.

Known Properties

Colour: Deep orange-amber, darker than typical evaporite crusts

Odour: Faint, sharp

Solubility: Fully miscible with water

Experiment Results

Measure pH: Strongly acidic, pH around 1.0–1.3

Measure density: Notably dense, around 1.4 g/mL

Test electrical conductivity: Very high

Add a magnet: No attraction

Add potassium thiocyanate solution: Turns deep blood-red almost immediately

Add sodium hydroxide solution: Forms an abundant reddish-brown gelatinous precipitate

Heat a small sample gently: Darkens slightly; no boiling yet

Experiments

Measure pH

Measure density

Test electrical conductivity

Add a magnet

Add potassium thiocyanate solution

Add sodium hydroxide solution

Heat a small sample gently

AI Behaviour

Reveal remaining candidates after narrowing: No

Socratic questions: Yes

Difficulty: Standard (15-16 years old)

Teacher Notes

REAL SAMPLE GROUP, assign an actual iron(III) chloride solution, concentrated (suggested ≈ 0.8 M FeCl_3 , adjust to your available stock). This is the most concentrated of the five real groups: 55 Cancri e is the hottest and most volcanically active planet in the set, so framing the deposit as heavily evaporated (hence concentrated) is the pedagogical link to real evaporite geology (comparable to how Earth's hottest, driest basins, e.g. the Dead Sea, Death Valley, produce the most concentrated salt deposits). In the real lab, this group should obtain the largest mass of iron oxide/hydroxide precipitate of the five real groups, the intended 'best source' result for the final conference ranking.

Real-science notes (for you, not necessarily for students in this form): 55 Cancri e is a genuine, well-studied hot rocky exoplanet, ~ 41 light-years away, orbiting the Sun-like star 55 Cancri A in 17.7 hours. JWST (Hu et al. 2024, Nature) found real evidence of a volatile-rich secondary atmosphere (not vaporised rock), most likely CO/CO₂ from magma outgassing, an actual 2024 measurement, not speculation. Reported dayside temperatures vary substantially between observations (~ 1700 – 2400 °C), itself a real and interesting finding (possibly patchy cloud cover or volcanic activity), a good discussion point on measurement uncertainty in real science. There is no reported detection of chlorine, iron, or any 'evaporite basin' on 55 Cancri e, the deposit and rover mission are an invented framing built on real planetary conditions (extreme heat, volcanism, a genuine confirmed atmosphere), not an actual finding.

Sources: Hu et al. 2024, Nature ([nature.com/articles/s41586-024-07432-x](https://www.nature.com/articles/s41586-024-07432-x)); NASA Exoplanet Catalog (science.nasa.gov/exoplanet-catalog/55-cancri-e).

Cross-subject link: Biology selective pressure for this planet, extreme heat and volcanic toxicity on a world with a permanent lava ocean.